

Notes on Regular Epimorphisms and the Monadicity of the Pullback Functor in an Exact Category

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These notes are based on the work of Sobral and Tholen [ST92], with some ideas drawn from Janelidze and Tholen [JT91]. Our goal is to prove the following theorem.

Theorem. In an exact category, if $p : E \rightarrow B$ is a regular epi, then p^* is a monadic functor.

We recall that a right adjoint is monadic if the comparison functor $K : (\mathcal{A} \downarrow B) \rightarrow (\mathcal{A} \downarrow E)^T$ is an equivalence of categories, where $T = p^*p_!$ is the monad associated to p .

$$\begin{array}{ccc}
 (\mathcal{A} \downarrow B) & \xrightarrow{K} & (\mathcal{A} \downarrow E)^T \\
 \uparrow p_! \quad \downarrow p^* & \nearrow F^T & \\
 (\mathcal{A} \downarrow E) & \nwarrow U^T &
 \end{array}$$

Let $\mathcal{A}$ be a category with pullbacks and let $p : E \rightarrow B$ be a morphism in $\mathcal{A}$. Let $(C, \gamma; \xi)$ be an object in $(\mathcal{A} \downarrow E)^T$ with $\xi : T(C) \rightarrow C$ an algebra.

Consider the pullback in $\mathcal{A}$.

$$\begin{array}{ccc}
 E \times_B C & \xrightarrow{\text{proj}_{2,C}} & C \\
 \text{proj}_{1,C} \downarrow & \nearrow \gamma & \downarrow p\gamma \\
 E & \xrightarrow{p} & B
 \end{array}$$

Thus, we have the span in $\mathcal{A}$:

$$\begin{array}{ccc}
 & E \times_B C & \\
 \text{proj}_{2,C} \swarrow & & \searrow \xi \\
 C & & C
 \end{array}$$

Since ξ is also a morphism in $(\mathcal{A} \downarrow E)$, we have that $\gamma\xi = \text{proj}_{1,C}$

$$\begin{array}{ccc}
 E \times_B C & \xrightarrow{\xi} & C \\
 \text{proj}_{1,C} \searrow & & \swarrow \gamma \\
 & E &
 \end{array}$$

In the following for the p^* -image of a morphism f in $(\mathcal{A} \downarrow B)$ we will use both notation $p^*(f)$ and $1_E \times_C f$.

Lemma 1. *Let $\mathcal{A}$ be a finitely complete category and let $p : E \rightarrow B$ be a morphism in $\mathcal{A}$. For any object $(C, \gamma; \xi)$ in $(\mathcal{A} \downarrow E)^T$, with T the monad associated to p , the morphism ξ is the coequalizer of $(1_E \times_B \xi, 1_E \times_B \text{proj}_{2,C})$.*

Proof. Let $(C, \gamma; \xi)$ be an object in $(\mathcal{A} \downarrow E)^T$. First we aim to show that $\xi(1_E \times_B \xi) = \xi(1_E \times_B \text{proj}_{2,C})$. This holds because ξ is an algebra for the monad $T = (p^*p_!, p^*\epsilon p_!, \eta)$. Observe that the multiplication in an object (C, γ) is

$$p^*\epsilon p_!(C, \gamma) = p^*(\epsilon_{(C, p\gamma)}) = p^*(\text{proj}_{2,C}) = 1_E \times_B \text{proj}_{2,C}$$

Then we have the algebra property

$$\begin{array}{ccc} E \times_B (E \times_B C) & \xrightarrow{1_E \times_B \xi} & E \times_B C \\ \downarrow 1_E \times_B \text{proj}_{2,C} & & \downarrow \xi \\ E \times_B C & \xrightarrow{\xi} & C \end{array}$$

$$\xi(1_E \times_B \text{proj}_{2,C}) = \xi(1_E \times_B \xi)$$

It remains to prove the universal property of the coequalizer. First we observe that ξ is a split epi, in fact $\xi\eta(C, \gamma) = 1_C$. Thus ξ is the coequalizer of $(1_{E \times_B C}, \eta_{(C, \gamma)}\xi)$. We aim to prove that ξ is also the coequalizer of $(1_E \times_B \xi, 1_E \times_B \text{proj}_{2,C})$. We already know that the universal property holds for ξ with $(1_{E \times_B C}, \eta_{(C, \gamma)}\xi)$, so it suffices to show that there exist a morphism $t : E \times_B C \rightarrow E \times_B (E \times_B C)$ such that

- $(1_E \times_B \xi)t = \eta_{(C, \gamma)}\xi$
- $(1_E \times_B \text{proj}_{2,C})t = 1_{E \times_B C}$

Then given a morphism f such that $f(1_E \times_B \text{proj}_{2,C}) = f(1_E \times_B \xi)$, we have, composing with t ,

$$\begin{aligned} f(1_E \times_B \text{proj}_{2,C})t &= f(1_E \times_B \xi)t \\ \implies f1_{E \times_B C} &= f\eta_{(C, \gamma)}\xi \end{aligned}$$

By the universal property for ξ with $(1_{E \times_B C}, \eta_{(C, \gamma)}\xi)$ we obtain the universal property for ξ with $(1_E \times_B \xi, 1_E \times_B \text{proj}_{2,C})$.

$$\begin{array}{ccccc} & & E \times_B C & & \\ & \swarrow t & \downarrow 1_E \times_B C & \downarrow \eta_{(C, \gamma)}\xi & \\ E \times_B (E \times_B C) & \xrightarrow[1_E \times_B \text{proj}_{2,C}]{1_E \times_B \xi} & E \times_B C & \xrightarrow{f} & F \\ & & \downarrow \xi & \nearrow \exists! g & \\ & & C & & \end{array}$$

We now have to prove that such a morphism t actually exists. Define $t = \langle \text{proj}_{1,C}, 1_{E \times_B C} \rangle$. We aim to prove that this morphism satisfies the two conditions above, ie the following diagram of pairs commutes.

$$\begin{array}{ccc}
 & E \times_B C & \\
 \langle \text{proj}_{1,C}, 1_{E \times_B C} \rangle \swarrow & \downarrow 1_{E \times_B C} \quad \eta_{(C,\gamma)} \xi & \\
 E \times_B (E \times_B C) & \xrightarrow[1_{E \times_B \text{proj}_{2,C}}]{1_E \times_B \xi} & E \times_B C
 \end{array}$$

In order to prove that we use the jointly monic pair $(\text{proj}_{1,C}, \text{proj}_{2,C})$. Observe that the following compositions hold.

- $\text{proj}_{2,C}(1_E \times_B \xi) \langle \text{proj}_{1,C}, 1_{E \times_B C} \rangle = \xi \text{proj}_{2,E \times_B C} \langle \text{proj}_{1,C}, 1_{E \times_B C} \rangle = \xi 1_{E \times_B C} = \xi$
- $\text{proj}_{2,C} \eta_{(C,\gamma)} \xi = \text{proj}_{2,C} \langle \gamma, 1_C \rangle \xi = 1_C \xi = \xi$
- $\text{proj}_{2,C}(1_E \times_B \text{proj}_{2,C}) \langle \text{proj}_{1,C}, 1_{E \times_B C} \rangle = \text{proj}_{2,C} \text{proj}_{2,E \times_B C} \langle \text{proj}_{1,C}, 1_{E \times_B C} \rangle = \text{proj}_{2,C} 1_{E \times_B C} = \text{proj}_{2,C}$
- $\text{proj}_{2,C} 1_{E \times_B C} = \text{proj}_{2,C}$
- $\text{proj}_{1,C}(1_E \times_B \xi) \langle \text{proj}_{1,C}, 1_{E \times_B C} \rangle = \text{proj}_{1,E \times_B C} \langle \text{proj}_{1,C}, 1_{E \times_B C} \rangle = \text{proj}_{1,C}$
- $\text{proj}_{1,C} \eta_{(C,\gamma)} \xi = \text{proj}_{1,C} \langle \gamma, 1_C \rangle \xi = \gamma \xi = \text{proj}_{1,C}$
- $\text{proj}_{1,C}(1_E \times_B \text{proj}_{2,C}) \langle \text{proj}_{1,C}, 1_{E \times_B C} \rangle = \text{proj}_{1,E \times_B C} \langle \text{proj}_{1,C}, 1_{E \times_B C} \rangle = \text{proj}_{1,C}$
- $\text{proj}_{1,C} 1_{E \times_B C} = \text{proj}_{1,C}$.

Thus $t = \langle \text{proj}_{1,C}, 1_{E \times_B C} \rangle$ satisfies the conditions required. $\square$

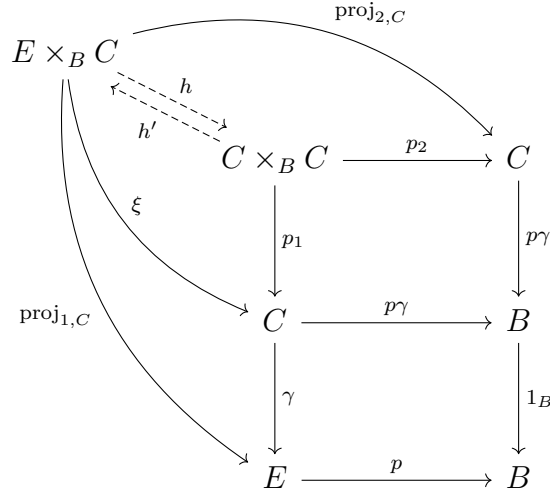
Lemma 2. *Let $\mathcal{A}$ be a finitely complete category and let $p : E \rightarrow B$ be a morphism in $\mathcal{A}$. For any object $(C, \gamma; \xi)$ in $(\mathcal{A} \downarrow E)^T$, with T the monad associated to p , the pair $(\text{proj}_{2,C}, \xi)$ is an equivalence relation in $\mathcal{A}$*

Proof. First we show that $(\text{proj}_{2,C}, \xi)$ is an internal relation. We need to show that $\langle \text{proj}_{2,C}, \xi \rangle$ is a mono. Consider the following diagram of solid arrows¹. It commutes, in fact

- $(p\gamma)\text{proj}_{2,C} = 1_B(p\gamma)\text{proj}_{2,C} = p\text{proj}_{1,C} = p\gamma\xi$
- $1_B(p\gamma)p_2 = 1_B(p\gamma)p_1 = p\gamma p_1$

¹[ST92] assumes the existence of coequalizers, which is not the case in our setting. However, this is not a problem: at this point, we can use the morphism $p\gamma$ instead of the coequalizer of $(\text{proj}_{2,C}, \xi)$, as it serves our purpose. Then, *a posteriori*, this proposition ensures, under the assumption of exact category, that the pair $(\text{proj}_{2,C}, \xi)$ has a coequalizer, which we will make use of.

Thus we can consider the morphism h' induced by the universal property of the pullback from p_2 and γp_1 , and h similarly induced from $\text{proj}_{2,C}$ and ξ .



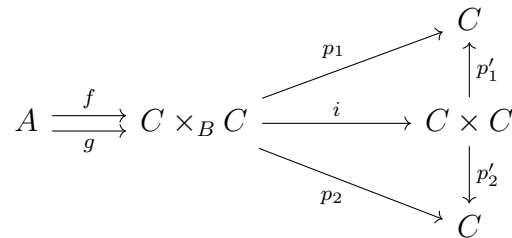
In order to prove that $\langle \text{proj}_{2,C}, \xi \rangle$ is a mono, we first need to show that h is a mono. Since $(\text{proj}_{1,C}, \text{proj}_{2,C})$ is a jointly monic pair, by the following compositions

- $\text{proj}_{1,C} h' h = \gamma p_1 h = \gamma \xi = \text{proj}_{1,C} = \text{proj}_{1,C} 1_{E \times_B C}$
- $\text{proj}_{2,C} h' h = p_2 h = \text{proj}_{2,C} = \text{proj}_{2,C} 1_{E \times_B C}$

we have that $h' h = 1_{E \times_B C}$. This implies that h is a mono, in fact suppose that f and g are two morphisms such that $h f = h g$, then we have

$$\begin{aligned}
 h f &= h g \\
 \implies h' h f &= h' h g \\
 \implies 1_{E \times_B C} f &= 1_{E \times_B C} g \\
 \implies f &= g
 \end{aligned}$$

Consider now the canonical morphism $i : C \times_B C \rightarrow C \times C$ induced by the universal property of the product. We have that i is a mono, in fact if f and g are two morphisms such that $i f = i g$, then we have



- $i f = i g \implies p'_1 i f = p'_1 i g \implies p_1 f = p_1 g$

- $if = ig \implies p'_2 if = p'_2 ig \implies p_2 f = p_2 g$

Since (p_1, p_2) is a jointly monic pair, we have that $f = g$. Thus, i is a mono. Since i and h are both monos, we have that ih is a mono.

By the following commutative diagram

$$\begin{array}{ccccc}
 & & \xi & & \\
 & \nearrow & & \searrow & \\
 E \times_B C & \xrightarrow{h} & C \times_B C & \xrightarrow{i} & C \times C \\
 & \searrow & \nearrow p_1 & \nwarrow p_2 & \\
 & & C & & C \\
 & & \uparrow p'_1 & & \downarrow p'_2 \\
 & & C & & C \\
 & \nwarrow \text{proj}_{2,C} & & \nearrow & \\
 & & C & &
 \end{array}$$

we conclude that $ih = \langle \text{proj}_{2,C}, \xi \rangle$, so $\langle \text{proj}_{2,C}, \xi \rangle$ is a mono.

We proved that $(\text{proj}_{2,C}, \xi)$ is an internal relation, now we need to show that it is an equivalence, ie it is reflexive, symmetric and transitive.

- Claim: $(\text{proj}_{2,C}, \xi)$ is reflexive.

Called η the unit of the monad, we have that $\langle \text{proj}_{2,C}, \xi \rangle \eta_{(C,\gamma)} = \langle 1_C, 1_C \rangle$. In fact $\xi \eta_{(C,\gamma)} = 1_C$ and by the definition of the unit of the monad, we have that $\text{proj}_{2,C} \eta_{(C,\gamma)} = \text{proj}_{2,C} \langle \gamma, 1_C \rangle = 1_C$.

$$\begin{array}{ccccc}
 & & C & & \\
 & \swarrow 1_C & \downarrow \eta_{(C,\gamma)} & \searrow 1_C & \\
 C & \xleftarrow{\text{proj}_{2,C}} & E \times_B C & \xrightarrow{\xi} & C
 \end{array}$$

- Claim: $(\text{proj}_{2,C}, \xi)$ is symmetric.

Consider the following diagram, where $\text{proj}_{1,E \times_B C}$ and $\text{proj}_{2,E \times_B C}$ are the projections of $E \times_B (E \times_B C)$. Let l be the morphism induced by the universal property of the pullback from $1_{E \times_B C}$ and $\gamma \text{proj}_{2,C}$, in particular it is the unique such that

$$\gamma \text{proj}_{2,C} = \text{proj}_{1,E \times_B C} l = \text{proj}_{1,C} (1_E \times \text{proj}_{2,C}) l$$

and

$$1_{E \times_B C} = \text{proj}_{2,E \times_B C} l$$

Consider the pullback (???)

$$\begin{array}{ccc} (E \times_B C) \times_C (E \times_B C) & \longrightarrow & E \times_B C \\ \downarrow & & \downarrow \xi \\ E \times_B C & \xrightarrow{\text{proj}_{2,C}} & C \end{array}$$

□

In an exact category each equivalence relation is a kernel pair and by (CITE) has a coequalizer. Thus in an exact category $(\text{proj}_{2,C}, \xi)$, by *Lemma 2* has a coequalizer. Let $q : C \rightarrow Q$ be such coequalizer. Observe that, considering $\text{proj}_{2,C}$ and ξ as morphism in $(\mathcal{A} \downarrow B)$, then q is a morphism in $(\mathcal{A} \downarrow B)$ as well by the universal property. So, called δ the morphism induced by the universal property of the coequalizer from $\text{proj}_{2,C}$ and ξ , we have that q with codomain (Q, δ) is also the coequalizer of the pair in $(\mathcal{A} \downarrow B)$.

$$\begin{array}{ccccc} E \times_B C & \xrightleftharpoons[\text{proj}_{2,C}]{\xi} & C & \xrightarrow{q} & Q \\ & \searrow p\text{proj}_{1,C} & \downarrow p\gamma & \swarrow \delta & \\ & & B & & \end{array}$$

Consider the comparison functor $K : (\mathcal{A} \downarrow B) \rightarrow (\mathcal{A} \downarrow E)^T$ defined as $K(D, \zeta) = (p^*(D, \zeta), p^*(\epsilon_{(D, \zeta)}))$, where ϵ is the counit of the adjunction, and $\epsilon_{(D, \zeta)}$ is defined as the projection from $E \times_B D$ to D of the pullback of p along ζ .

In our context of exact category, turns out that the comparison functor K has a left adjoint $L : (\mathcal{A} \downarrow E)^T \rightarrow (\mathcal{A} \downarrow B)$ defined as $L(C, \gamma; \xi) = (Q, \delta)$, where (Q, δ) is the codomain of coequalizer of the pair $(\text{proj}_{2,C}, \xi)$ in $(\mathcal{A} \downarrow B)$.

$$\begin{array}{ccc} (\mathcal{A} \downarrow B) & \xrightleftharpoons[L]{K} & (\mathcal{A} \downarrow E)^T \\ \uparrow p! & \nearrow F^T & \\ \downarrow p^* & \nwarrow U^T & \\ (\mathcal{A} \downarrow E) & & \end{array}$$

Let us introduce the counit ρ and counit α of the adjunction $K \dashv L$. For each object (D, ζ) in $(\mathcal{A} \downarrow B)$ the counit is a morphism of the kind $\rho_{(D, \zeta)} : LK(D, \zeta) \rightarrow (D, \zeta)$. First we compute $K(D, \zeta)$.

$$K(D, \zeta) = (E \times_B D, \text{proj}_{1,D}; p^*(\text{proj}_{2,D}))$$

$$\begin{array}{ccc} E \times_B D & \xrightarrow{\text{proj}_{2,D}} & D \\ \downarrow \text{proj}_{1,D} & & \downarrow \zeta \\ E & \xrightarrow{p} & B \end{array}$$

Then, call $q' : E \times_B D \rightarrow Q'$ the coequalizer of the equivalence relation $(\text{proj}_{2, E \times_B D}, p^*(\text{proj}_{2, D}))$, as done above we make it a coequalizer in $(\mathcal{A} \downarrow B)$ via the universal property. We obtain $q' : (E \times_B D, p\text{proj}_{1, D}) \rightarrow (Q', \delta')$. Then compute $LK(D, \zeta)$. By definition of the pullback functor, the top left square in the diagram below commutes. Thus we can define, as shown in the diagram, the morphism $\rho_{(D, \zeta)}$ by the universal property of the coequalizer q' .

$$\begin{array}{ccccc}
 E \times_B (E \times_B D) & \xrightarrow[p^*(\text{proj}_{2, D})]{\text{proj}_{2, E \times_B D}} & E \times_B D & \xrightarrow{q'} & Q' \\
 \downarrow \text{proj}_{2, E \times_B D} & & \downarrow \text{proj}_{2, D} & \nearrow \rho_{(D, \zeta)} & \\
 E \times_B D & \xrightarrow{\text{proj}_{2, D}} & D & & \\
 \downarrow \text{proj}_{1, D} & \nearrow \text{proj}_{1, D} & \downarrow \zeta & \searrow \delta' & \\
 E & \xrightarrow{p} & B & &
 \end{array} \tag{1}$$

Observe that δ' was also defined as the morphism induced by the universal property of the coequalizer, considering the object $E \times_B D$ as an object in $(\mathcal{A} \downarrow B)$ via $p\text{proj}_{1, D}$. Thus $\rho_{(D, \zeta)}$ is a morphism in $(\mathcal{A} \downarrow B)$, ie $\zeta \rho_{(D, \zeta)} = \delta'$.

For each object $(C, \gamma; \xi)$ in $(\mathcal{A} \downarrow E)^T$ the unit is a morphism of the kind $\alpha_{(C, \gamma; \xi)} : (C, \gamma; \xi) \rightarrow KL(C, \gamma; \xi)$. We compute $KL(C, \gamma; \xi)$. Let $q : C \rightarrow Q$ be the coequalizer of the equivalence relation $(\text{proj}_{2, C}, \xi)$ in $\mathcal{A}$. By the universal property of the coequalizer, we have that q is a morphism in $(\mathcal{A} \downarrow B)$ via $p\gamma$. Thus, we can consider the coequalizer of the equivalence relation $(\text{proj}_{2, C}, \xi)$ in $(\mathcal{A} \downarrow B)$, which we denote by $q : (C, p\gamma) \rightarrow (Q, \delta)$. By functoriality, we have that $p^*(q)p^*(\xi) = p^*(q)p^*(\text{proj}_{2, C})$. Moreover, it holds that ξ is the coequalizer of $(p^*(\xi), p^*(\text{proj}_{2, C}))$ by *Lemma 1*. Thus we can define $\alpha_{(C, \gamma; \xi)}$ as the morphism induced by the universal property of the coequalizer q , as done in the diagram below. One can check that $\alpha_{(C, \gamma; \xi)}$ is a morphism in $(\mathcal{A} \downarrow E)^T$.

$$\begin{array}{ccccc}
 E \times_B (E \times_B C) & \xrightarrow[p^*(\text{proj}_{2, C})]{p^*(\xi)} & E \times_B C & \xrightarrow{p^*(q)} & E \times_B Q \\
 \downarrow \text{proj}_{2, E \times_B C} & & \downarrow \text{proj}_{2, C} & \nearrow \alpha_{(C, \gamma; \xi)} & \downarrow \text{proj}_{2, Q} \\
 E \times_B C & \xrightarrow[\text{proj}_{2, C}]{\xi} & C & \xrightarrow{q} & Q \\
 \downarrow \text{proj}_{1, C} & \nearrow \gamma & \downarrow p\gamma & \searrow \delta & \\
 E & \xrightarrow{p} & B & &
 \end{array} \tag{2}$$

One can show that ρ and α are natural transformations, and that they satisfy the universal property of the counit / unit. This follows from the universal property of the coequalizer of the equivalence relation considered.

Our goal is to show that p^* is a monadic functor, ie that the comparison functor K is an equivalence of categories. We prove that it is an adjoint equivalence, which is the same. In order to do so, we aim to prove that the unit α and the counit ρ of the adjunction $L \dashv K$ are pointwise isomorphisms.

Proposition 1. *In an exact category if $p : E \rightarrow B$ is a regular epi, then the counit ρ of the adjunction $L \dashv K$ is a pointwise isomorphism.*

Proof. We aim to prove that for each object (D, ζ) in $(\mathcal{A} \downarrow B)$, the morphism $\rho_{(D, \zeta)} : LK(D, \zeta) \rightarrow (D, \zeta)$ is an isomorphism. Consider the diagram (1) of the definition of the counit $\rho_{(D, \zeta)}$. An exact category is also regular, in particular regular epi are pullback stable. If p is a regular epi, then $\text{proj}_{2,D}$ is a regular epi as well. The top left square is a pullback by (CITE), thus $(p^*(\text{proj}_{2,D}), \text{proj}_{2,E \times_B D})$ is the kernel pair of $\text{proj}_{2,D}$. Since $\text{proj}_{2,D}$ is a regular epi, then, by (CITE), it is the coequalizer of its kernel pair $(p^*(\text{proj}_{2,D}), \text{proj}_{2,E \times_B D})$. We have that q' and $\text{proj}_{2,D}$ are coequalizers of the same pair, thus the morphism $\rho_{(D, \zeta)}$ defined by the universal property of the coequalizer q is actually an isomorphism. $\square$

Proposition 2. *In an exact category the unit α of the adjunction $L \dashv K$ is a pointwise isomorphism.*

Proof. We aim to prove that for each object (C, γ, ξ) in $(\mathcal{A} \downarrow B)$, the morphism $\rho_{(D, \zeta)} : LK(D, \zeta) \rightarrow (D, \zeta)$ is an isomorphism. Consider the diagram (2). An exact category is also regular, in particular regular epi are pullback stable. Since q is a regular epi, then $p^*(q)$ is a regular epi as well. Moreover, since q is the coequalizer of $(\xi, \text{proj}_{2,C})$ which is a kernel pair, then by (CITE) $(\xi, \text{proj}_{2,C})$ is the kernel pair of q . Then, since p^* is a right adjoint, it preserves kernel pairs, thus $(p^*(\xi), p^*(\text{proj}_{2,C}))$ is the kernel pair of $p^*(q)$. We already proved that $p^*(q)$ is a regular epi, then by (CITE) it is the coequalizer of its kernel pair $(p^*(\xi), p^*(\text{proj}_{2,C}))$. By *Lemma 1* also ξ is the coequalizer of $(p^*(\xi), p^*(\text{proj}_{2,C}))$. We have that $p^*(q)$ and ξ are coequalizers of the same pair, thus the morphism $\alpha_{(C, \gamma; \xi)}$ defined by the universal property of the coequalizer ξ is actually an isomorphism. $\square$

Finally, we have the main result we were aiming to prove.

Theorem 1. *In an exact category, if $p : E \rightarrow B$ is a regular epi, then p^* is a monadic functor.*

Proof. The exactness of the category implies that the comparison functor $K : (\mathcal{A} \downarrow B) \rightarrow (\mathcal{A} \downarrow E)^T$ has a left adjoint $L : (\mathcal{A} \downarrow E)^T \rightarrow (\mathcal{A} \downarrow B)$, as shown. Since p is a regular epi, the unit α and the counit ρ of the adjunction $L \dashv K$ are pointwise isomorphisms, by *Proposition 1* and *Proposition 2*. Thus, $L \dashv K$ is an adjoint equivalence. This implies that K is an equivalence of categories. So $p^* : (\mathcal{A} \downarrow B) \rightarrow (\mathcal{A} \downarrow E)$ is a monadic functor. $\square$

References

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